MSC THESIS DEFENSE

Data-Driven Cognitive Phenotyping in Acquired Brain Injury

Do routine neuropsychological data reveal cognitive subtypes?

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WHY THIS MATTERS

The clinical problem: heterogeneity

- ABI can disrupt attention, memory, language, executive function, orientation and visuospatial cognition.
- Impairment does not follow the same pattern in every patient.
- Diagnosis and injury severity are useful categories — but they do not capture cognitive heterogeneity.
- Incomplete assessments are common, especially among more impaired patients.

Research question

Can data-driven clustering recover clinically meaningful cognitive organisation when missing data are handled explicitly — rather than ignored?

POSITIONING

Research gap and contribution

Common limitation in prior work	This dissertation
Samples often below 200 participants	17,406 assessments · 7,285 patients
Complete-case analysis	Ten imputation strategies compared
Limited algorithmic sensitivity testing	Hyperparameter, seed and missingness checks
Predetermined number of clusters	Density-based clustering (HDBSCAN)
Limited clinical validation	Association with clinical-unit assignment

PRE-REGISTRATION

Four hypotheses, objective criteria

H1	**Are stable clusters recoverable?**	Clusters separate cleanly, leave few cases unassigned, and form at least two groups `Silhouette > 0.40 · noise < 30% · ≥ 2 clusters`
H2	**Robust to imputation choice?**	Cluster assignments agree strongly whichever imputation method is used `Mean pairwise ARI > 0.50 (fixed-reference)`
H3	**Are tiers clinically relevant?**	Severity tier relates to clinical unit, even after accounting for diagnosis χ^2 test · Cramér's V > 0.10 · CMH adjust for diagnosis
H4	**Does domain aggregation help?**	Grouping variables into domains gives cleaner, more stable separation `Higher silhouette · lower VIF & condition number`

THE DATA

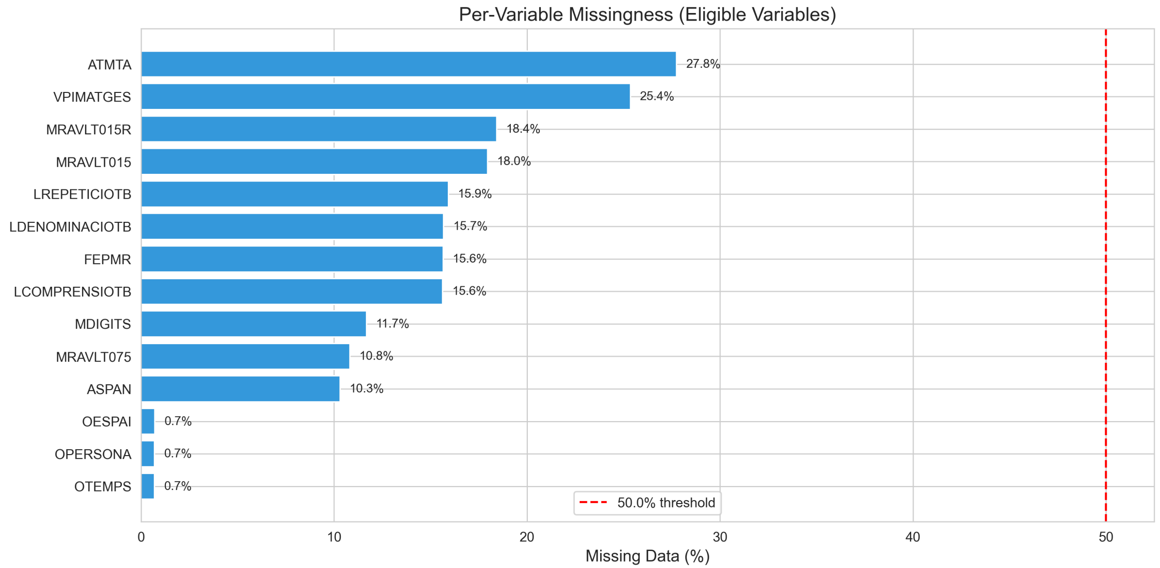
A large but incomplete cohort

17,406
assessments

7,285
patients

14
variables

6
cognitive domains



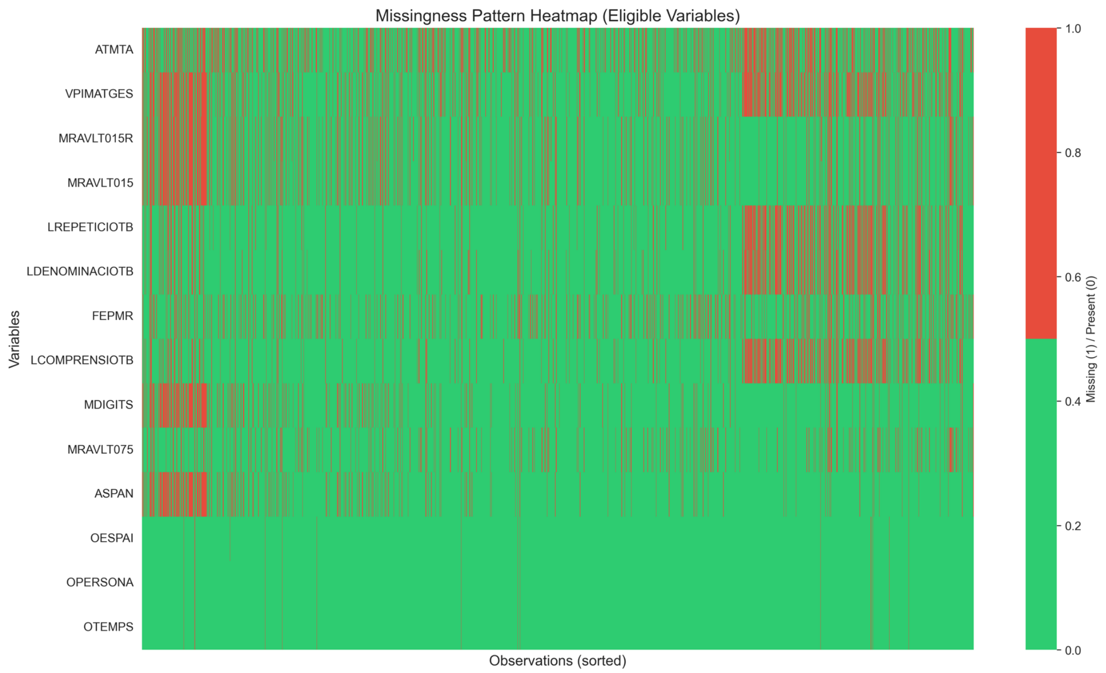
53.1%

of rows are complete across all 14 variables

- Complete-case analysis would discard ≈ 47% of assessments.
- Missingness is non-random — it tracks the testing process.
- Imputation is needed to retain partially observed patients.

METHOD RATIONALE

Why imputation is necessary



Missingness reflects the clinical testing process

- Fatigue or reduced motivation
- Cognitive overload or task aversion
- Disorientation or motor limitations
- Limited tolerance for long batteries
- Clinician selects a more suitable test

Dropping incomplete rows preferentially removes the most impaired patients — so missingness must be managed and tested, not hidden.

METHODS

Analysis pipeline

Impute	Scale	Aggregate	UMAP	HDBSCAN
10 strategies simple → deep learning	orient scores — higher = better	14 variables → 6 domains	manifold embedding	density clustering + sensitivity

Fixed-reference design (H2)

One MICE solution freezes the scaler, UMAP embedding and HDBSCAN partition; every other imputation is then projected through this fixed structure — its missing values placed on the same map and into the same clusters.

✓ **Tests** the same patient lands in the same tier whichever imputation filled the gaps — geometry held fixed.

✗ **Not tested** that re-fitting the whole pipeline per dataset would rediscover the same cluster hierarchy.

UMAP

Uniform Manifold Approximation & Projection · McInnes, Healy & Melville, 2018

- Builds a fuzzy k-NN graph of neighbourhood structure.
- Balances local and global geometry.
- Reduces dimensionality before density clustering.

Caveat: separation in UMAP space can be optimistic — it must be backed by sensitivity analyses.

FORMAL DEFINITION

$$p_{j|i} = \exp\Big(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\Big), \qquad \sum_j p_{j|i} = \log_2 k$$

$$p_{ij} = p_{j|i} + p_{i|j} - p_{j|i}\, p_{i|j}$$

$$q_{ij} = \left(1 + a\,\|y_i - y_j\|_2^{2b}\right)^{-1}$$

$$\mathcal{C} = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}} + (1 - p_{ij}) \log \frac{1 - p_{ij}}{1 - q_{ij}}$$

HDBSCAN

Hierarchical Density-Based Spatial Clustering of Applications with Noise · Campello, Moulavi & Sander, 2013

- Needs no predefined number of clusters.
- Finds regions dense across density levels.
- Labels uncertain points as noise, not forced.

The minimum cluster size is the main knob; everything else follows from the density hierarchy.

FORMAL DEFINITION

$$\mathrm{core}_k(x) = d\big(x,\, N_k(x)\big)$$

$$d_{\mathrm{mreach}}(a,b) = \max\{\mathrm{core}_k(a),\, \mathrm{core}_k(b),\, d(a,b)\}$$

$$S(C) = \sum_{x \in C} \big(\lambda_{\max}(x) - \lambda_{\mathrm{birth}}(C)\big), \qquad \lambda = \frac{1}{\mathrm{dist}}$$

$$\widehat{\mathcal{P}} = \arg\max_{\mathcal{P}} \; \textstyle\sum_{C \in \mathcal{P}} S(C) \quad (\text{non-nested})$$

Ten imputation strategies

Simple → model-based

1. Mean imputation
2. Median imputation
3. Predictive mean matching
4. Expectation–maximization
5. MICE (reference)

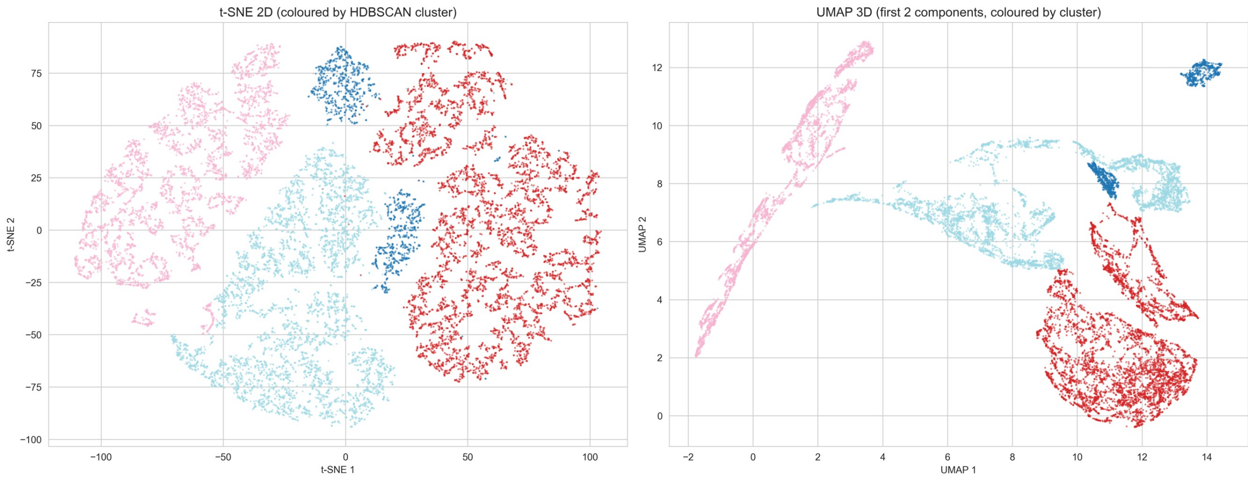
Donor / matrix / deep learning

6. K-nearest neighbours
7. MissForest
8. SoftImpute
9. Non-negative matrix factorization
10. Deep generative approaches

Agreement across this full range is what makes the recovered tiers credible — not an artefact of one preprocessing choice. Settings in Appendix A.

RESULT · H1

Three cognitive severity tiers emerge



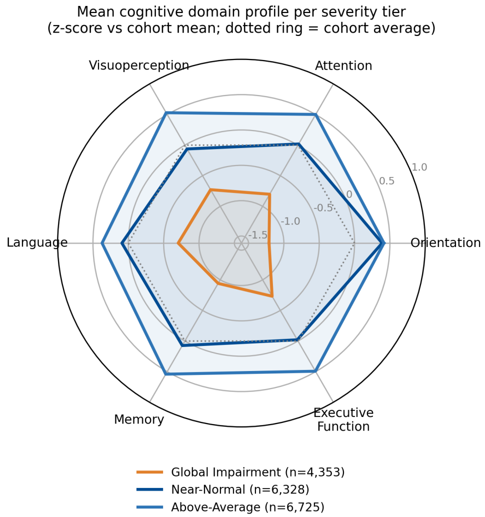
Above-Average 6,725
above cohort mean

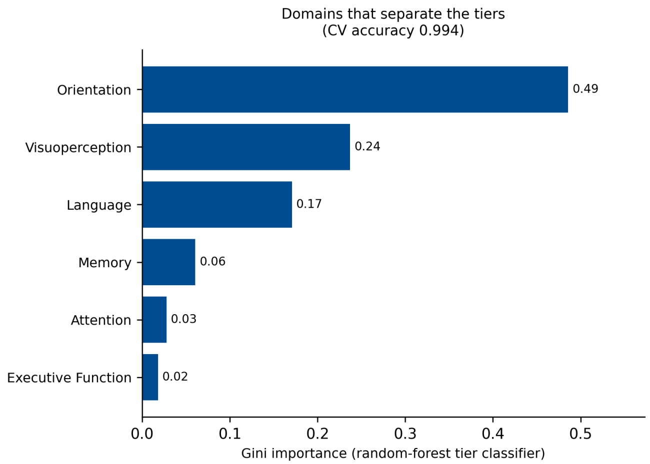
Near-Normal 6,328
close to the average

Global Impairment 4,353
below cohort mean

All 10 imputations cleared pre-registered thresholds · MICE reference silhouette ≈ 0.53 · k = 3. Criteria met, but the tiers are not interpreted as essential categories.

Continuum evidence: level, not shape





PC1

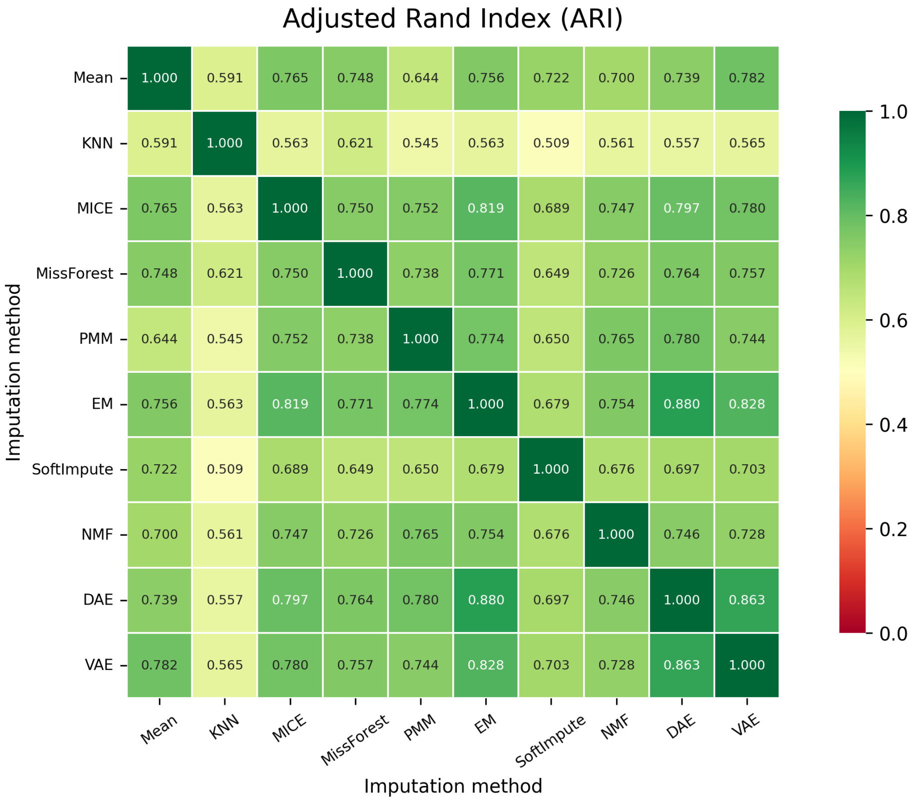
≈ 56%

of variance — one dominant axis

- Radar profiles are roughly concentric.
- All domains move in the same direction.
- All domains load positively on PC1.
- Signal = global severity, not dissociation.

RESULT · H2

Robust across imputation methods



Pairwise ARI across all ten imputation methods

0.71

mean pairwise ARI (95% CI 0.70–0.72)

44 / 45

method pairs survive Holm correction

> 86%

assessments with high-confidence consensus

Scope: point assignments are robust under a fixed scaler, embedding and partition — not full invariance of independently re-fit pipelines.

RESULT · H2

Complete-case analysis changes the answer

	MICE reference	Complete-case
Assessments retained	100%	53.1%
Tiers recovered	3	2
Silhouette	≈ 0.53	≈ 0.26
ARI vs. reference (shared rows)	—	≈ 0.63

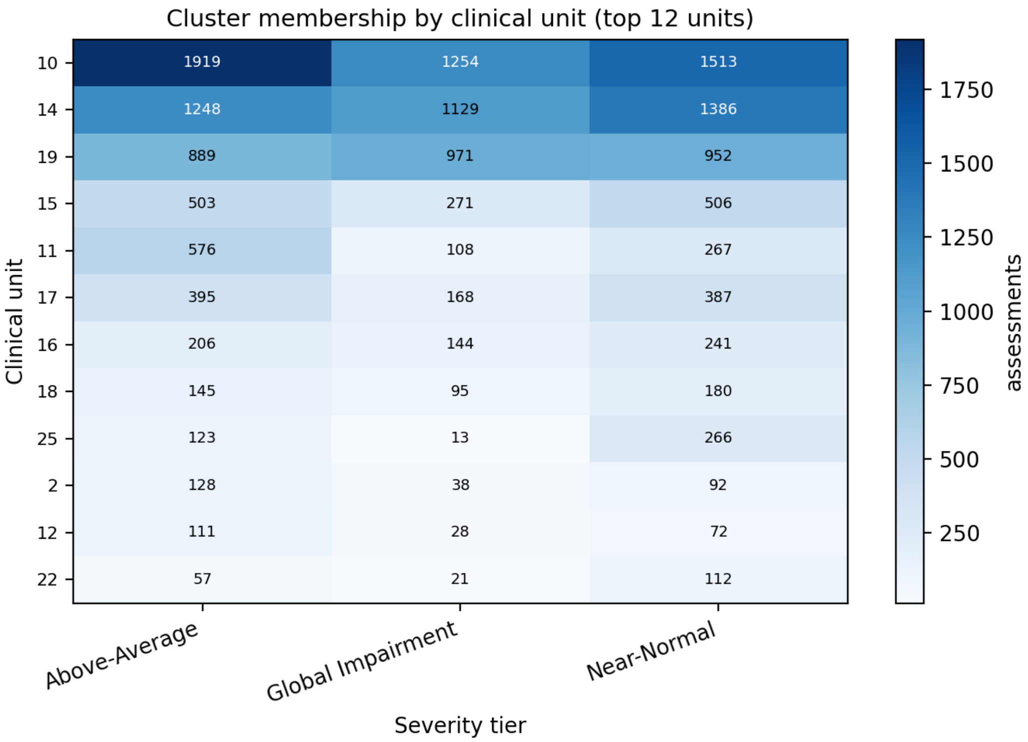
Deleting cases is not a weaker version of the same analysis —

it changes the partition.

The baseline collapses three tiers to two and halves the silhouette. How missing data is handled becomes part of the model's substantive output.

RESULT · H3

Tiers track clinical unit assignment



$\chi^2 = 911$ $p < 10^{-156}$

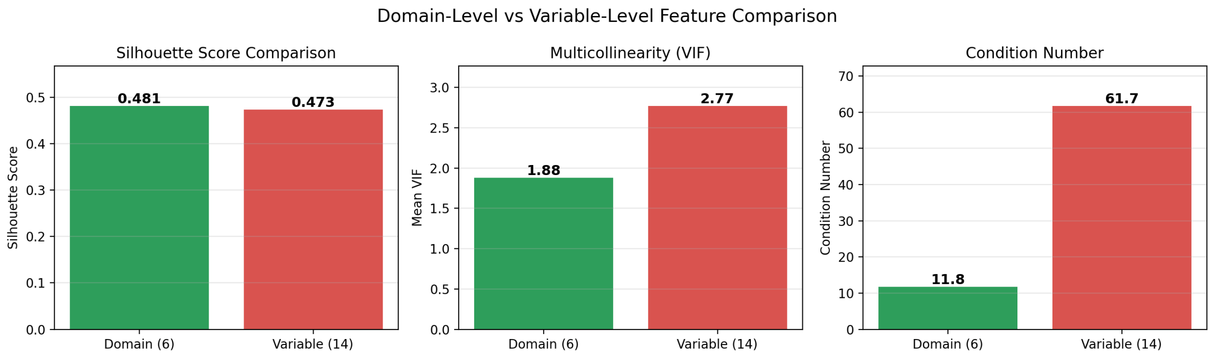
0.16

Cramér's V (pre-registered threshold 0.10)

- Cochran–Mantel–Haenszel: holds after adjusting for diagnosis.
- Severity tier adds information beyond the diagnostic label.
- Informs rehabilitation routing — it does not dictate placement.

RESULT · H4

Domain aggregation is more stable



Silhouette

~~0.473~~ → **0.481**

variables → domains

Mean VIF

~~2.77~~ → **1.88**

variables → domains

Condition number

~~61.7~~ → **11.8**

variables → domains

WHAT IT MEANS

A continuum, not a catalogue

Above-Average	Near-Normal	Global Impairment

less impaired — more impaired

- Subtypes would imply qualitatively different cognitive profiles.
- A gradient implies differences lie mostly along one severity axis.
- The tiers are practical bands extracted from that continuum.
- Domain-specific dissociations may exist — but they don't dominate here.
- Poor scaling, redundant or non-cognitive features, or complete-case selection can fake subtype structure.

IMPLICATIONS

Clinical and analytic utility

Clinical users

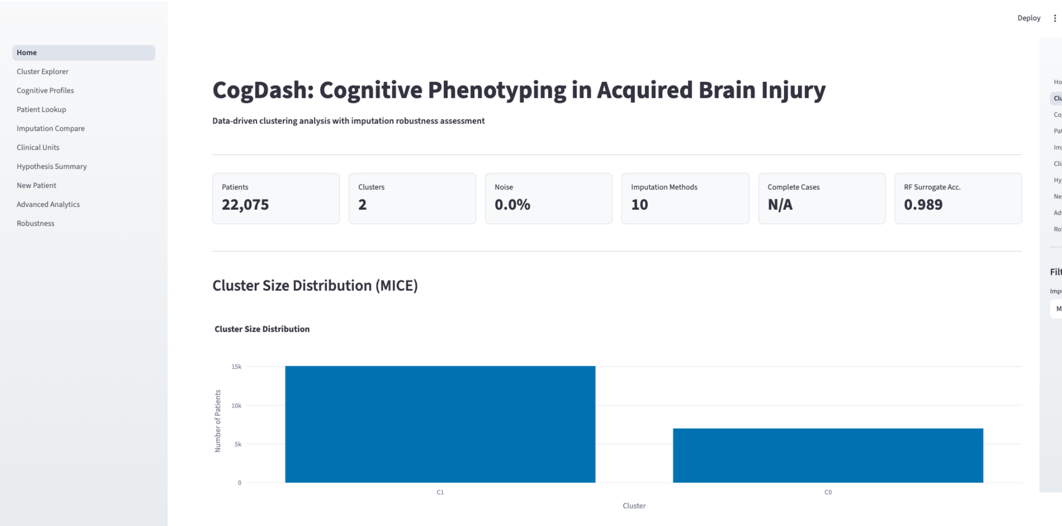
- Shared vocabulary for rehabilitation intensity.
- Confidence scores for borderline assignments.
- Information alongside diagnosis and judgement.

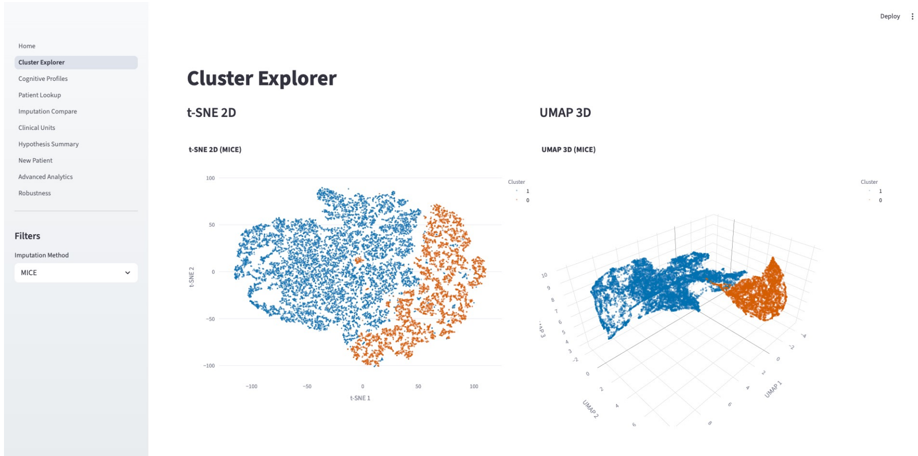
Analytic users

- Reproducible missingness-aware pipeline.
- Explicit, testable imputation decisions.
- Adaptable for linking cognition to outcomes.

Long-term contribution: make missing-data decisions explicit, then test their influence — rather than hiding them.

CogDash — a transparent decision-support tool





- Cluster explorer · patient lookup · new-patient classifier · hypothesis summaries.
- Built in Streamlit — auditable inputs and visible confidence.
- Supports collaboration; it does not automate clinical decisions.

CAVEATS

Limitations

Single site

One rehabilitation centre — external replication is required.

Limited longitudinal modelling

Repeated assessments hint at recovery, but no full trajectory model.

Missing-data assumptions

MAR is a working assumption; MNAR cannot be excluded.

No functional outcomes

Return-to-work, length-of-stay and independence were unavailable.

Embedding-based validation

UMAP-space silhouette may overstate separation.

Pipeline sensitivity

Aggregation + density clustering may favour a general severity axis.

IN SUMMARY

Three conclusions

1	**Empirical**	Routine post-ABI neuropsychological data carry a strong cognitive-severity gradient — ordered tiers, not discrete phenotypes.
2	**Methodological**	Imputation is a substantive modelling decision that can alter the recovered structure.
3	**Practical**	Domain-level severity tiers are interpretable and numerically stable — preferable to raw-variable clustering.

THANK YOU

Questions & discussion

A clinically useful severity continuum after acquired brain injury — more conservative than claiming discrete cognitive subtypes.

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APPENDIX

Backup slides for examiner questions

OVERVIEW

Roadmap

1. Clinical problem and research gap
2. Cohort, missingness, and methods
3. Four pre-registered hypothesis tests
4. Interpretation and practical utility
5. Limitations and conclusions

Imputation hyperparameters

Method	Hyperparameters
Mean	Single-pass arithmetic mean per variable.
EM	Multivariate normal model; convergence 1e−6; max 500 iterations.
MICE	Linear regression per variable; 10 iterations; single imputed dataset for clustering.
KNN	$k = 5$; inverse-distance weighting; Euclidean distance.
MissForest	100 trees per forest; up to 10 iterations; variables imputed by increasing missingness.
PMM	5 donors; 10 iterations; implemented within the MICE framework.
SoftImpute	Regularisation chosen by CV over 20 log-spaced values (1e−3 to 1e2); convergence 1e−5; max 200 iterations.
NMF	Rank $r = 5$ selected by grid search over {2, 3, 5, 7, 10}; multiplicative updates; max 500 iterations.
DAE	15-128-64-32 autoencoder; ReLU + batch norm; dropout 0.5/0.3; 20% pattern-aware masking; MSE on observed entries; Adam (lr 1e−3); early stopping; 5 runs averaged.
VAE	Same layer dimensions as DAE; latent dim 32; loss = MSE + β-weighted KL (cyclical annealing, β_{max} 0.1); 20 latent samples averaged.

H2 fixed-reference design

- Fit the reference scaler on MICE-imputed data.
- Fit one UMAP embedding from the MICE reference.
- Define the reference HDBSCAN partition.
- Transform each alternative imputed dataset through the frozen pipeline.
- Compare predicted labels using pairwise ARI.

Tests

Robustness of point assignments to imputed values.

Does not test

Full invariance of independently discovered structures.

HDBSCAN logic

- Estimate local density via core distance.
- Construct mutual-reachability distances.
- Build a minimum spanning tree.
- Extract a hierarchy of density-connected components.
- Select persistent clusters by stability.
- Label points outside persistent regions as noise.

Unlike DBSCAN, HDBSCAN does not reduce to a single fixed-ε DBSCAN solution — it integrates across density levels.

APPENDIX · D

Silhouette caveats

- UMAP is designed to emphasise neighbourhood structure.
- Silhouette measured in embedded space can overstate separation.
- Hyperparameter and seed selection make the chosen estimate optimistic.
- Seed stability, PCA and alternative representations provide supporting evidence.

Interpret silhouette as an internal pipeline-quality metric — not proof of natural categories.

Missing-not-at-random sensitivity

- MAR was the principal working assumption.
- MNAR mechanisms cannot be ruled out from observed values alone.
- Extreme-MNAR analyses suggest Global Impairment is comparatively robust.
- Boundaries between the less-impaired tiers are more sensitive.

APPENDIX · F

Clinical interpretation boundary

The severity tier DOES

- supplement diagnosis;
- support rehabilitation discussion;
- carry a confidence estimate;
- assist outcome modelling.

The severity tier does NOT

- diagnose a patient;
- prescribe treatment;
- determine clinical placement;
- replace professional judgement.

APPENDIX · G

Future work

- External replication across rehabilitation centres.
- Direct longitudinal modelling of recovery trajectories.
- Validation against return-to-work, length-of-stay and independence.
- Alternative model-based phenotyping (e.g. latent profile analysis).
- Prospective evaluation of CogDash in clinical workflows.
- Greater sensitivity to domain-specific dissociations.

How to read the metrics

Clustering quality & robustness

Silhouette

Cohesion vs. separation — how well a point fits its own tier vs. the nearest other.

−1 to 1; higher = cleaner. Pre-reg > 0.40 · observed ≈ 0.53

Adjusted Rand Index (ARI)

Agreement between two partitions, corrected for chance.

0 = chance, 1 = identical. Pre-reg > 0.50 · observed ≈ 0.71

Normalised Mutual Information

Shared information between two partitions, scaled 0–1.

0 = independent, 1 = identical · corroborates ARI

Consensus confidence

Share of assessments given a stable label across imputations.

> 86% high-confidence across methods

Association & conditioning

χ^2 statistic

Evidence that tier and clinical unit are associated — significance, not effect size; grows with N.

$\chi^2 = 911$ · $p < 10^{-156}$

Cramér's V

Effect size of that association, scaled 0–1.

0.1 small · 0.3 medium · 0.5 large. Pre-reg > 0.10 · observed 0.16

VIF (variance inflation factor)

Per-feature redundancy from collinearity.

1 = none; > 5–10 concerning. 2.80 → 1.90

Condition number

Global conditioning of the feature set ($\lambda_{max} / \lambda_{min}$ of the correlation matrix).

< 10 fine · 10–30 moderate · > 30 strong. 62 → 12